BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, KOHAT

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2023

PHYSICS

Past Paper 2023 • Paper Code: K1-IX-23-I

Retyped faithfully from the original board paper — same questions, no changes.

Section "A"

Time Allowed: 20 Minutes | **Maximum Marks:** 12 | **Paper Code:** Code A

Note: Q.1 Attempt all questions of section-A by filling the correct option i.e. A, B, C or D on the OMR RESPONSE SHEET. Cutting, erasing and over writing will not be awarded any marks.

Qi. In measurement 1.032 numbers of significant figures are .....

(A) 1 (B) 2

(C) 3 (D) 4

Qii. A train is moving at a speed of $10ms^{-1}$. It's speed expressed in kmh^{-1} is .....

(A) 36 kmh^{-1} (B) 40 kmh^{-1}

(C) 0.01 kmh^{-1} (D) 30 kmh^{-1}

Qiii. A weight of a body is 147N. Its mass will be ..... when g = $10ms^{-2}$.

(A) 1.47 kg (B) 14.7kg

(C) 47 kg (D) 1.51 kg

Qiv. The net torque acting on a rotating body with uniform speed is .....

(A) 1Nm (B) 2Nm

(C) 5Nm (D) 0Nm

Qv. The value of 'g' on the surface of Moon is .....

(A) 1.62 ms^{-2} (B) 10 ms^{-2}

(C) 20 ms^{-2} (D) zero

Qvi. The amount of work done in carrying a bag of mass 10kg to a height of 3.6 m is .....

(A) 36J (B) 3.6J

(C) 3600J (D) 360J

Qvii. Unit of Young's Modulus is .....

(A) Nm (B) Nm^{-2}

(C) Nm^{-1} (D) Nm^{2}

Qviii. How much heat energy is required to melt 250g of ice?.

(A) 84000J (B) 840000J

(C) 1344J (D) 134400J

Qix. The bad conductor of heat is .....

(A) Iron (B) copper

(C) air (D) aluminium

Qx. A force of 10N is making an angle of 30° with horizontal. It's horizontal component will be .....

(A) 4N (B) 5N

(C) 7N (D) 8.7N

Qxi. Rolling frication is ..... then sliding friction.

(A) greater (B) less

(C) same (D) half

Qxii. The relation between coefficient of linear expression and volume expansion is .....

(A) $\alpha = 3\beta$ (B) $\beta = 3\alpha$

(C) $\alpha = 3/\beta$ (D) $\beta = 3/\alpha$

Section "B"

Time Allowed: 2 Hours 40 Minutes | **Maximum Marks:** 32

QQ.2. Attempt any EIGHT parts. All parts carry equal marks.

i Define any four branches of physics?

ii How much time is required to change 44Ns momentum by a force of 40N?.

iii Describe a situation in which the speed of an object is constant while the velocity is not.

iv Find magnitude and direction of a force by its rectangular components.

v Calculate mass of earth by using law of gravitation.

vi A crane is capable of doing 1.50×10^{5} J of work in 10s. what is the power of crane in watts and hp (horse power).

vii Differentiate between boiling and evaporation.

viii Explain the reason of land breeze and sea breeze.

ix How is convection current useful in heating and ventilation system of buildings?

x Derive equation $2as = vf^2 - vi^2$ graphically.

xi A car has kinetic energy K. what will be the effect on its kinetic energy if it's velocity is reduced to half.

Section "C"

Maximum Marks: 21

Note: Note: Attempt any THREE of the following questions. All questions carry equal marks.

QQ.3.

a What is equilibrium?. What are the conditions of equilibrium.

b To open a door force of 15N is applied at 30° to the horizontal, find the horizontal and vertical components.

QQ.4.

a Two bodies of masses m_1 and m_2 are attached through a string passing over a frictionless Polly. Derive an expression for tension in string (T) and acceleration (a).

b What is centripetal force?. Also show that $Fc = mv^2/r$.

QQ.5.

a State Archimedes principle. How can the density of an object be determined using Archimedes Principle?.

b An 80cm long, 1.0 mm diameter steel guitar string must be tightened to a tension of 2000N by turning the tuning screws. By how much is the string stretched?.

QQ.6.

a What is meant by thermal expansion . also prove that $Lt = L_o(1+\alpha\Delta T)$.

b How much heat is required to increase the temperature of 0.5kg of water (with specific heat capacity of 4190 $Jkg^{-1}K^{-1}$) from 10 °C to 65 °C ?.